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DEVELOPMENTS IN CHINESE MEDICINE FROM THE SONG THROUGH THE QING

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Introduction

When practitioners of Chinese medicine look to pre-modern literature for insight and inspiration, we find different kinds of information in the writings of each era. In terms of practical application, medical literature can be broadly divided into the writings of the last twelve hundred years, and the entirety of what preceded it. We typically refer to the corpus of these earlier writings to deepen our understanding of the larger concepts of Chinese medicine. We may read the *Yellow Emperor's Inner Classic* (*Huangdi neijing* 黃帝內經, Eastern Han 東漢, 25–225 CE), hereafter *Inner Classic*, for a fresh contextual perspective on a difficult case, but as a rule we do not expect to find a specific treatment protocol in the *Inner Classic* that will solve a particular clinical conundrum.

In the literature from the Song dynasty (960–1279) onwards, however, we find precisely this kind of clinical detail. We see how the core ideas presented in the *Inner Classic* and the other seminal texts of our literature were interpreted and extrapolated upon in very concrete and tangible ways. This provides us with a multiplicity of approaches adapted to address a vast range of circumstances and individual presentations. The *Discussion of Cold Damage* (*Shanghan lun* 傷寒論, c. 220), a seminal treatise on the progression and treatment of febrile disease, by Zhang Ji 張機 (stylename Zhongjing 仲景, 150–219) is an apt illustration of this phenomenon (Chapter 8 in this volume).[2] Although now regarded as a cornerstone of medical literature of the first few centuries of the common era, the *Discussion of Cold Damage* became much more influential in the Song, and virtually every clinical perspective on the text in use today was developed in the Song or later.[3] At least in terms of its clinical relevance today, the *Discussion of Cold Damage* is a Song document. Chinese medical writings of the past thousand years may be read as a multi-faceted articulation of the canonical literature in clinical practice. This chapter will provide an overview of some of the most important milestones in the development of that body of knowledge and the circumstances that produced them.

The Song dynasty

In recent years, the Song dynasty has been recognised in Anglophone scholarship as a pivotal period in the development of Chinese medicine (Hinrichs 2003; Leung 2003a, b;

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Scheid 2007; Goldschmidt 2009). By then, the foundations of medical theory were well established, and as Asaf Goldschmidt has observed, it was 'during the Song that all the intellectual habits of what is now thought of as Chinese medicine assumed their mature form' (Goldschmidt 2009: 2). Definitive versions of canonical texts were printed, cosmological and physiological principles were more thoroughly integrated into pharmacology and 'medicine' (*yi* 醫) – defined as the administration of drug therapies and, to a lesser extent, acupuncture – eclipsed other ritualistic and shamanistic forms of healing (Hinrichs 2003). The number of medicinals recorded in the *materia medica* doubled during the Northern Song (960–1127) (Shang Zhijun 2002: 34–57). During the Song, the role of the scholar-physician also emerged as a definitive social feature, and medical practice gained an unprecedented level of legitimacy. A number of factors were responsible for these changes.

Governmental oversight

The Song dynasty witnessed unprecedented government involvement in the regulation and distribution of medicine, which was at least partially a political response to the epidemics sweeping the country during this time. This, and a genuine interest in medicine among imperial rulers, prompted the development of official prescription references such as the *Formulary of the Pharmacy Service for Benefiting the People in an Era of Great Peace* (*Taiping huimin hejiju fang* 太平惠民和劑局方, 1107),[4] which were published and distributed by the court. Such formularies contained fixed prescriptions designed to treat specific disorders, making them accessible to the masses, which had the effect of radically reducing the demand for specialists.

Governmental involvement in medical practice gradually extended to the credentialling of practitioners and the production of texts under the Bureau for Revising Medical Texts (*jiaozheng yishu ju* 校正醫書局) (Liang Jun 1995: 97–8). Its interest in both practitioners and the manner in which medicine was practised ultimately gave rise to governmentally run dispensaries, the first of which was established in 1076.

The flowering of Cold Damage

In terms of clinical practice, the flowering of interest in *Discussion of Cold Damage* was among the most significant theoretical developments of the Song. Again, at least partially in response to the epidemics that plagued this time, the Bureau for Revising Medical Texts published ten primary medical texts. These included an authoritative edition of the *Discussion of Cold Damage*, and two other texts concerning Cold Damage theory, as well as three more texts directly relevant to it, testifying to the influence of this approach to treatment (Goldschmidt 2009: 91–5). During this time, concerns with Cold Damage diseases ultimately overshadowed interests in other medical pursuits. Texts such as Pang Anshi's 龐安時 (stylename Anchang 安常, 1042–99) *General Discussion of the Disease of Cold Damage* (*Shanghan zongbing lun* 傷寒總病論, 1100) and Zhu Gong's 朱肱 (stylename Yizhong 翼中, fl. 1088) *Lifesaving Book that Categorises the Presentations of Cold Damage* (*Shanghan leizheng huoren shu* 傷寒類證活人書, 1108) both systematised Cold Damage theory and made it more easily accessible for laypeople.

Perhaps the most significant Cold Damage text of this period was Cheng Wuji's 成無己 (1066–1156) *Annotation and Explanation of the Discussion of Cold Damage* (*Zhujie shanghan lun* 注解傷寒論, 1144). Cheng explained the course of Cold Damage disorders in the context of the classical doctrines linking the channels and viscera.[5] This innovative synthesis became part of a medical orthodoxy that physicians in subsequent eras would actively attempt to dismantle (Goldschmidt 2009: 169; Scheid 2013).

Scholarly medicine

Prodigious population growth during the Song created a need for jobs among the educated (Hymes 1987; Scheid 2007: 37). Beginning in the Northern Song, governmental endorsement of medicine as a profession helped to legitimise medical practice and produced career opportunities for the gentry (Boyanton 2015: 171–220). Fan Zhongyan's 范仲淹 (stylename Xiwen 希文, 989–1052) admonition that '[if] one cannot become a good minister one [should strive] to become a good physician' is indicative of the new-found status of medicine during this time (Scheid 2007: 41). By the Southern Song (1127–1279), literati were taking up the profession in large numbers and physicians began to be regarded as elite exponents of cultivated ideas (Ibid.: 35).

Book printing facilitated intellectual exchange both among physicians and between physicians and scholars, making medicine an increasingly scholarly pursuit. In this environment, Daoist and Buddhist ideals also became deeply integrated into Confucian thought. An important characteristic of this Neo-Confucianism was the principle of personal cultivation through the investigation of things (*gewu* 格物), and the place of humanity in them, for the betterment of society as a whole.[6] Scholars steeped in this sensibility who also had an interest in medicine came to be referred to as *ruyi* 儒醫 (scholar-physicians). In the Southern Song (1127–1279), the term came to denote not only the degree of a physician's intellectual understanding but also his own level of psychological and spiritual cultivation, and embodied the conviction that both of these qualities were essential to effective medical practice (Ibid.: 39–41).

Jin and Yuan dynasties

The elite ranks of scholarly physicians, or at least those claiming that status, blossomed in the Jin 金 and Yuan 元 dynasties (1115–1368). Their interest in adapting the classical medical literature to contemporary medical concerns was a driving force in the evolution of Chinese medicine. Diagnostic focus shifted away from single pathogenic factors such as wind or cold, and descriptions of specific medical disorders (*bing* 病), and towards a new focus on the dynamics of *qi* transformation and how the breakdown of this transformation expressed itself in distinctive presentations (*zheng* 證; Ibid.: 43).

Scholar-physicians of the Jin-Yuan period further challenged the Song orientation towards established formulæ and globally defined disorders by focusing on local disease processes (Boyanton 2015: 164 ff.). Although government involvement in healthcare fostered greater access to medical information among the general population in the form of treatment manuals, and even previously secret hereditary prescriptions were made public, scholar-physicians directly attacked these formulary approaches as a deeply degraded form of medicine. From their perspective, administering fixed formulæ based on identification of a patient's disease was insufficient. Prescriptions should be carefully tailored to the patient and the patient's relationship to their immediate environment.

Systematisation in diagnostics and pharmacology

The organisation and systematisation of diagnostic parameters that had begun in the Later Han with the *Inner Classic* assumed a more mature form in the Jin-Yuan. The paediatrician Qianyi 錢乙 (stylename Zhongyang 仲陽, 1032–1113) and Kou Zongshi 寇宗奭 (fl. 1111–1117) had already begun to articulate the eight necessities (*bayao* 八要) for effective

treatment. Synthesising these parameters, Zhang Yuansu 張元素 (stylename Jiegu 潔古, fl. 1151–1254) systematised pattern presentation of the viscera and receptacles (*zangfu* 臟腑) and articulated relationships between the viscera and receptacles with the channels and networks (*jingluo* 經絡), and the six *qi*. It was Zhang Yuansu who expressed the concepts of root diseases (*benbing* 本病), those pathodynamics that are at the root of a problem, and branch diseases (*biaobing* 標病), the typically symptomatic expression of an illness. He also defined what today is considered eight-principle differentiation (*bagang* 八綱), that any presentation should be described in terms of whether it is *yin* or *yang*, excess or deficient, hot or cold, and whether it originates from inside or outside the body (Buck 2009).

A similar consolidation of concepts occurred in the field of pharmacology where there was an ongoing effort to describe drugs in a more comprehensive and coherent manner. Wang Haogu's 王好古 (stylename Haizang 海藏, thirteenth century) *Materia Medica for Decoctions* (*Tangye bencao* 湯液本草, 1298) was especially significant in its articulation of a pharmacology of systematic correspondence, presenting a system for treating specific conditions based on drug qualities derived from Five Phase (or Five Agent) theory (Unschuld 1986: 111).[7] This trend culminated in Zhu Zhenheng's 朱震亨 (stylename Danxi 丹溪, 1281–1358) identification of multiple primary drug qualities that rationalised the apparent contradictions between theoretical and empirical effects of drugs (Ibid.: 187).

The four freat masters of the Jin-Yuan

Of the many scholar-physicians active during the Jin-Yuan period, four in particular advanced a sequence of related ideas that significantly contributed to the development of medical theory during this time. These were Liu Wansu 劉完素 (stylename Shouzhen 守真 c. 1120–1200), Zhang Congzheng 張從正 (stylename Zihe 子和, 1156–1228), Li Gao 李杲 (stylename Dongyuan 東垣, 1180–1251) and Zhu Zhenheng 朱震亨 (stylename Danxi 丹溪, 1281–1358).

Liu Wanusu

For Liu Wansu, all pathogenic factors, regardless of their origin, eventually manifest as pathogenic fire. Therefore, cool and cold medicinals are indicated to treat nearly all conditions. One of Liu's characteristic strategies was to drain heart fire while nourishing kidney water, an idea that would be developed by subsequent thinkers, most notably Zhu Zhenheng. Framing this core idea was a belief that diagnosis and treatment must take into account environmental and climatic conditions, as well as the patient's constitutional predispositions. This would become a defining perspective of all four masters of the Jin-Yuan and a general principle of the time.

Zhang Congzheng

Like Liu Wansu, Zhang Congzheng's primary focus was on the accumulation of exogenous pathogenic factors and improper food intake, and he downplayed the role of internally generated pathologies due to the emotions or constitutional weakness. Zhang's fundamental approach involved the removal of these pathogens through diaphoresis, emesis and purgation, expanding the definition of these strategies as originally outlined in *Discussion of Cold Damage* and employing a wider range of therapies that included acupuncture, moxibustion, massage and exercise. Even more vigorously averse to tonifying medicinals than Liu Wansu, Zhang Congzheng preferred to employ dietary stratagems rather than pharmaceuticals when supplementation was indicated.

Li Gao

Li founded his approach on the *Inner Classic's* assertion that the spleen and stomach were the basis of the body's *qi*, and a belief that they could be compromised due to improper food intake, overexertion and emotional disorders (Welden 2015). Although Xu Shuwei 許叔微 (stylename Heke 知可, 1079–1154) had already emphasised the importance of the stomach during the Song, Li made this insight the centrepiece of his own thinking while integrating Zhang Yuansu's discrimination of viscera and receptacle syndrome patterns.

Where Liu Wansu and Zhang Congzheng's overall approach to therapy stressed the elimination of excess pathogenic factors, those of Li Gao and Zhu Zhenheng focused more on replenishing metabolic insufficiencies. Later scholars came to believe that this shift in emphasis reflected the deteriorating political circumstances during the latter part of Jin-Yuan period. According to Xu Dachun 徐大椿 (stylename Lingtai 靈胎, 1693–1771):

> When in the final years of the Song dynasty the Chinese territory fell into the hands of the enemy [referring to the northerners], the head of state was weak and his ministers failed to show any strength. At the same time, Zhang Yuansu, Li Gao, and others, in establishing their prescriptions, emphasised supplementation of the central palace.
>
> *(Xu Lingtai yixue quanshu: 140)*[8]

A weak central government was analogous to weak digestive function. Strengthening the central government/digestion was the key to both political and individual health (Weldon 2015).

The spleen stomach current became one of the most influential perspectives in subsequent Chinese medical history. Li Gao's most mature work, his *Discussion of the Spleen and Stomach* (*Piwei lun* 脾胃論, thirteenth century), was very influential on later writers.[9] Wang Haogu, Li Shizhen, Ye Gui (see below) and others all acknowledged the centrality of the spleen and stomach even as they incorporated a broader range of concepts into their medical understanding.[10]

Zhu Zhenheng

The last of the four great masters, Zhu Zhenheng is generally regarded as representing the pinnacle of Jin-Yuan medical thought. His ideas reflect a synthesis of previous medical currents produced by a consummately cultivated philosopher-physician. Charlotte Furth has detailed the influence of Zhu's thought and practice of Zhu Xi's 朱熹 (1130–1200) Neo-Confucian philosophy of 'investigating things and extending knowledge' (*gewu zhizhi* 格物致知),[11] a realist philosophical approach to the external, physical world that sought to understand its basic principles and humanity's role in them (Furth 2006). Zhu Zhenheng was a prominent exemplar of this principle, which helped to propel and legitimise medical innovation from the Song to the end of the imperial period (Unschuld 1985: 195–6). As the most outspoken critic of formula books and the simplified Cold Damage therapeutics of his time, Zhu Zhenheng was an icon of the scholar-physician's intellectual agenda.

Zhu Zhenheng's reading of the three preceding Jin-Yuan currents led him to posit that the flaring of 'ministerial fire' (*xianghuo* 相火), the fiery aspect of *qi* that actually drove the physiological activity of the body, was the central cause of disease (Simonis 2010). Based on his reading of the *Inner Classic*, he understood ministerial fire as an active *yang* that gave rise to human desires and physiological functions. When ministerial fire became pathological, it was due to excessive longings and emotions, particularly of a sexual nature, impairing the

yin and blood. The nourishment of kidney *yin* was the primary means of controlling the ensuing symptoms of fire. This strategy has its basis in the Five Phase paradigm where water is the primary means of mitigating fire along the control (*ke* 克) cycle. Notwithstanding his emphasis on *yin*-nourishing therapies, Zhu regarded the support of the spleen and stomach as a central component of an overall *yin* nourishing strategy.[12]

Zhu Zhenheng's most important ideas are collected in *Danxi's Essential Methods (Danxi xinfa practitioner of the lowest rank*. 丹溪心法, 1347), and in a companion volume, *Extra Treatises based on Investigation and Inquiry* (*Gezhi yulun* 格致餘論, 1347), the title of which reflects his Neo-Confucian orientation.[13]

The Ming dynasty

By the Ming dynasty (1368–1644), all famous doctors were considered scholar-physicians, though the profession was still so poorly regulated that it was quite easy to make the claim that one was a physician, and so it remained a 'Minor Way' (*xiao dao* 小道) (Scheid 2007:45). Nevertheless, medicine had become a well-established means of social advancement. The family history of Li Shizhen 李時珍 (1518–93), among the most iconic figures in Chinese medical history, is emblematic of such a progression. Li's grandfather was an itinerant bell doctor (*lingyi* 鈴醫), a medical practitioner of the lowest rank. Li's father, Li Yuechi 李月池, passed the central state exams, elevating the family's social status considerably, though he remained a physician who became famous for his success in treating patients during a local epidemic.[14] These improved circumstances afforded Li Shizhen himself access to the world of the literati and court physicians making an especially broad range of medical information available to him.

Consolidation of medical knowledge

Li Shizhen

Li Shizhen 李時珍 (stylename Binhu 瀕湖, 1518–93) is among the most iconic figures in Chinese medical history. Each of his three surviving works constituted a definitive contribution to Chinese medicine, particularly with regard to the systematisation of knowledge within its specific field. Paul Unschuld describes Li's *Comprehensive Materia Medica* (*Bencao gangmu* 本草綱目, 1578) as the 'qualitative and quantitative climax in the development of the *materia medica* literature, expanding the genre from a work on drugs to a comprehensive and detailed encyclopaedia of medicine, pharmaceutics, minerology, metallurgy, botany and zoology' (Unschuld 1986: 163). Carla Nappi articulates Li Shizhen as a naturalist, and the *Comprehensive Materia Medica* as a seminal work of naturalist science (Nappi 2009). This magnum opus reflects a comprehensive familiarity with the medical literature and drug use throughout China at that time. In fact, the primary criticism of the work at the time of its publication was its very size, which made it unwieldy and expensive. The organisation of the book further refined the existing classification systems of the *materia medica* literature. Its title alludes to Zhu Xi's *Comprehensive Mirror to Aid in Government* (*Zizhi tongqian gangmu* 資治通鑑綱目), reflecting Li's Neo-Confucian sensibilities (Unschuld 1986: 151). Li's work is divided into individual monographs called *gang* 綱 (mainstays) and the criteria for the internal arrangement within the monographs are referred to as *mu* 目 (a hierarchical catalogue).[15] Li's works on pulse diagnosis, *Pulse Studies of the Lakeside Master* (*Binhu maixue* 瀕湖脈學, 1564) and on

the eight Extraordinary Vessels, *Exposition on the Eight Extraordinary Vessels* (*Qijing bamai kao* 奇經八脈考, 1577), have been similarly influential.[16]

Li's *Pulse Studies of the Lakeside Master* organised and codified pulse diagnosis into the twenty-seven pulses that are the basis for Chinese medical pulse diagnosis today.[17] Finally, his *Exposition on the Eight Extraordinary Vessels* synthesised and extrapolated upon previous ideas concerning the Extraordinary Vessels, particularly with regard to herbal medicine.[18] Li's integration of Daoist inner elixir (*neidan* 内丹) principles of internal cultivation into medical practice in this text is representative of the widespread interest in such practices during the Ming.[19]

Zhang Jiebin

Zhang Jiebin 張介賓 (stylename Jingyue 景岳, fl. 1562–1639) was an erstwhile soldier who developed the use of military terminology in medical treatment. He presented a perspective that was a counterpoint to the premise of Liu Wansu and Zhu Zhenheng that an excess of *yang* was the primary source of illness. For Zhang, *yang* was *qi* and *yin* was form, and since one could never have too much *qi*, one could never have too much *yang*. He is remembered as the foremost developer of the Life Gate (*mingmen* 命門) principle of thought and practice. He understood the Life Gate to be the manifestation of primordial *yin* and *yang* (*yuan yinyang* 元陰陽), residing between the kidneys.[20] As such, he stressed the enrichment of genuine or true *yin* (*zhen yin* 真陰). Although the Life Gate was the repository of former heaven (*xiantian* 先天) *yin* and *yang* endowed before birth, he recommended its replenishment through the latter heaven (or postnatal) essence (*houtian jing* 後天精) provided by the nourishment and digestive processes of the spleen and stomach.[21]

Regardless of whether it was the True *Yin* and True *Yang* (*zhenyang* 真陽) that appeared to be compromised, for Zhang Jiebin the problem was invariably rooted in a debilitation of the true *yin*. He, therefore, prioritised the nourishment of postnatal essence (*jing* 精) and blood (*xue* 血) in his treatment strategies, often through the support of spleen and stomach function. In this, his approach to treatment reflects a synthesis of spleen-, stomach- and *yin*-nourishing theories that flowered during the Jin-Yuan period. Zhang's *Categorisation of the Classic* (*Leijing* 類經, 1624) is among the most coherent, and systematically organised of the extant commentaries on the *Inner Classic*. It and Zhang's other most important writings are anthologised in his *Complete Works of Jingyue* (*Jingyue quanshu* 景岳全書, 1624).

Qing dynasty

The Qing (清) dynasty (1644–1911) was marked by a political and ideological conservatism that spawned the evidential scholarship (*kaozheng* 考證) movement, a re-examination of classical sources based on the principle that they were a more reliable basis for the resolution of the problems of the present than more modern ideas (Unschuld 1985: 193; Scheid 2007: 48). Its proponents argued that these earlier texts had not yet been corrupted by the Daoist and Buddhist principles characteristic of Song Neo-Confucianism. The introduction of Western science during this time almost certainly fuelled the evidential scholarship movement by providing multiple models of enquiry that in many cases had much greater predictive accuracy than the theory-laden structures of the past. In medicine, this played out as a search for the true understanding of classical sources, a quest that was tethered to the idea that wisdom must be uncovered through observation coupled with introspection.

Those involved in evidential scholarship not only directed medical interest away from the sage writings of antiquity, but also focused more on the experience of individual physicians

in their local environs. This trend centred on physicians in Jiangnan, particularly in and around Suzhou, fostering the invention of a southern medical tradition that was uniquely suited to the local *qi* (*tuqi* 土氣), which endowed the local inhabitants with a weaker (*rouruo* 柔弱) constitution than their northern counterparts (Hanson 1998).

Once again, new approaches to medical learning fostered a plurality of ideas tied to individual scholars. Yet, because none of these currents had become so predominant as to achieve the force of doctrine, they were ultimately identified as part of the problem (Unschuld 1985: 197). This plurality inspired efforts in the community of writers to redefine medical orthodoxy in texts such as the *Essential Readings from the [Orthodox] Medical Lineage* (*Yizong bidu* 醫宗必讀, 1637), and *Golden Mirror of the Medical Lineage* (*Yizong jinjian* 醫宗金鑑, 1739–42), which strove to identify medicine in terms of the canonical literature.[22] The vast range of individual approaches to learning was nevertheless legitimised by its representation in the state-sponsored anthology of Chinese knowledge, the *Complete Collection of the Four Treasuries* (*Siku quanshu* 四庫全書, 1773). A number of physicians, however, distinguished themselves as synthesisers of these two intellectual trends.

Synthesis

Ye Gui

Ye Gui 葉桂 (stylename Tianshi 天士, 1667–1746) was the most influential physician of the Qing period (1644–1912). In addition to his major contributions to the field of Warm Disease theory and practice, Ye was renowned as a virtuoso synthesiser and integrator of virtually all of the medical theories that preceded him, masterfully adapting a vast array of approaches and techniques to the demands of the moment. Ye himself is credited with writing very little and the clinical orientation of Ye's contribution is evident in the case records attributed to him that were collected by his students.

Ye's case records, particularly his *Case Record Guide to Clinical Presentation* (*Linzheng zhinan yi'an* 臨証指南醫案, 1746), became immensely popular after his death. Despite the fact that he was not of the scholarly class, Ye's own family strove to position him not as an innovator but as an exemplar of the scholar-physicians steeped in the literature of antiquity (Hanson 1997: 248).

Among the most significant of Ye's contributions to Chinese medical practice was his explicit emphasis on treatment strategies (*fa* 法) that could be flexibly adjusted, rather than on fixed formulæ. This privileging of methodology over diagnosis pioneered by Yu Chang 喻昌 (stylename Jiayan 嘉言, 1585–1664) was the logical extension of the scholarly concern with specific disease dynamics presenting in individual patients. Although it required a comprehensive understanding of the full spectrum of Chinese medical theories, it allowed for carefully tailored herbal prescribing. Despite this more generalised influence on medical practice, Ye Gui is best known in the West for his role in the development of Warm Disease theory.

Warm Disease

The role of heat as a cause of infectious disease had been a topic of discussion since the term Warm Disease (*wenbing* 溫病) was first mentioned in the *Inner Classic* as either a range of acute-onset febrile disorders or as a specific type of Cold Damage. Building on Liu Wansu's emphasis on fire as a more predominant pathogenic factor than cold, Wu Youxing 吳有性

(stylename Youke 又可, 1582–1652) posited that febrile epidemics (*wenyi* 溫疫) were due to pestilential *qi* (*liqi* 癘氣) that was hot in nature, establishing the conceptual foundation for Warm Disease therapeutics. From the seventeenth through the nineteenth centuries, Warm Disease theory developed as a methodology that was distinct from Cold Damage therapeutics (Hanson 2013: 204). Although the Warm Disease current is commonly positioned as a response to the failures of Cold Damage therapeutics, the concept of Warm Disease is both historically and conceptually a logical extension of the Cold Damage current, the principles of which are grounded in the *Discussion on Cold Damage* (Scheid 2013).[23] Similarly, Ye's development of new strategies for treating Warm Disease was undeniably innovative, and yet it is clear that his thinking was firmly rooted in the classically oriented sensibilities of the scholar-physician.

The one text that Ye Gui may well have written himself, the *Discourse on Warm Disease* (*Wenre lun* 溫熱論, 1792), systematised the progression of pathogenic heat into the body through four layers: defensive, *qi*, nutritive and blood. Within this theoretical structure, the text primarily concerned itself with the practicalities of differential diagnosis and treatment strategy. It is clearly written as a complement to the *Discussion on Cold Damage* and uses many of its ideas and therapies.[24] A significant portion of Ye's *Case Records Based on Clinical Presentation* is devoted to the treatment of Warm Disease patterns, providing concrete examples of how Ye made use of the therapeutic framework he laid out in his *Discourse*.

Xue Xue

Ye's contemporary rival, the scholar and poet Xue Xue 薛雪 (stylename Shengbai 生白, 1681–1770), posited a complementary approach to the four-layer model of disease progression that would form the other pillar of the theoretical framework of Warm Disease. Xue is credited with most thoroughly developing the Three Burner (*sanjiao* 三焦) model of illness that divided the location of pathogenic factors into upper, middle and lower parts of the body, roughly the thorax, epigastrium and lower abdomen. This model was particularly suited to damp heat (*shire* 濕熱) patterns, which he believed were not only different from cold patterns but also from Warm Diseases caused by heat alone (Liu 2001: 21). Thus, Xue's treatment approach reflects an abiding concern with the relative predominance of dampness or heat as it presents in a given location or warmer (Burner) in the body. Such patterns were then treated with particular attention to the stomach and spleen, which he believed were especially vulnerable to attack by damp heat (Qiu Peiran 1984: 278). Xue published a number of works concerning poetry and medicine. His most significant contribution to the Warm Disease current is his *Folio on Damp Heat Diseases* (*Shirebing pian* 濕熱病篇, 1852) edited by Wang Shixiong 王士雄 (stylename Mengying 孟英, 1808–67).[25]

Wu Tang

The person who brought the disparate streams of Warm Disease theory together was Wu Tang 吳瑭 (stylename Jutong 鞠通, 1758–1836). Building upon and substantially redefining the works of preceding Warm Disease physicians, Wu combined Xue Xue's Three Burner model dividing the torso into three regions, with Ye Gui's four-sector (*sifen* 四分) model identifying the relative depth of a pathogenic factor. In the process, he also linked each of the viscera and receptacles to a particular Burner, producing a finely detailed system for locating the progression of pathogenic heat within the body (Liu 2005: 23). His *Systematic Differentiation of Warm Disease* (*Wenbing tiaobian* 溫病條辨, 1798) has been immensely

influential in popularising the theory and treatment of Warm Disease. In many cases, the formulæ contained in this text are clearly extrapolations on the treatment strategies and herb combinations developed by Ye Gui. Wu was especially concerned with the loss of essence and believed that 'the words loss of storage of essence should be looked at flexibly and not explained exclusively an overindulgence in sex. Rather it is the case that all human activity can agitate and disturb the essence' (Liu 2001: 23).[26]

Conclusion

Even as a basis for critique, canonical literature has remained the impetus for the most significant developments in Chinese medical thinking over the last millennium. Attempts to understand such texts have produced a dynamic tension between the comparatively liberal or expansive interpretations exemplified by the four great masters of the Jin-Yuan, and relatively conservative or restrictive readings such as those of the Qing philologists. The creative interplay between these two impulses is such that one could not have occurred without the other. Each interpretation has produced a concrete and pragmatic response to the principles presented in the canonical literature. In this, Chinese medicine's perennial return to the classics may be best understood less as the search for an overarching truth than as an ongoing response to the challenge of developing effective therapies in the face of ever-changing circumstances.

Notes

1 Sadly, Charles 'Chip' Chace passed away while this volume was in production. Beloved by generations of Chinese medicine practitioners, Chip made a lasting impact in his published work, his training of many students, his kind humour and generous personality. It is fitting to share his chapter on the legacy of the Jin-Yuan masters, a legacy in which he himself took part. The editors wish to thank Volker Scheid for overseeing the final revisions.

2 Zhang Ji originally composed a single treatise known as the *Discourse on Cold Damage and Miscellaneous Diseases*. The earliest editor of this document, Wang Shuhe (王叔和, 201–280), subsequently divided the text into two parts, the *Discourse on Cold Damage* addressing infectious disease and the *Essential Prescriptions from the Golden Cabinet* (*Jingui yaolüe* 金匱要略) concerned with internal medicine. They have been considered as two separate texts ever since.

3 The modern clinician Huang Huang 黃煌 is among the most influential advocates of *shanghan*-based prescribing in both Asia and the West today. His approach is likely to be far more comprehensible to later *shanghan* thinkers than to Zhang Ji himself (Huang 2007). For an examination of the evolution of *shanghan* thinking, see Scheid (2013). For an overview of the range of clinical applications to the Cold Damage literature, see Chen and Zhang (1998). The best translation of the *Discussion of Cold Damage* to date is by Craig Mitchell, Feng Ye and Nigel Wiseman (Mitchell, Feng and Wiseman 1999).

4 *Taiping huimin heji ju fang* 太平惠民和劑局方 (*Formulary of the Pharmacy Service for Benefiting the People in the Taiping Era*) 1148, Liu Jingyuan 劉景源 (ed.) (1985), Beijing: Renmin weisheng chubanshe.

5 Ed's note: See Chapter 1 in this volume.

6 For a historical survey of the shifting concept of *gewu*, see Elman (2005: 5–9).

7 Ed: On the Five Phases (or Five Agents) (*wuxing* 五行) (see Introduction and Chapter 1 in this volume).

8 For a translation of Xu Dachun's *Yixue yuanliu lun* 醫學源流論 (Discourse on the Origin and Development of Medicine, 1764), see Unschuld (1990).

9 The *Treatise on the Spleen and Stomach* appeared relatively early in the wave of pre-modern Chinese medical texts translated into English beginning in the 1980s. Its appreciation of dietary and environmental aetiological factors, and its approach to modelling complex disease presentations have made it especially influential in the transmission of Chinese medical ideas into the West. Li's notion of '*yin* fire' (*yinhuo* 陰火) arising from a spleen stomach debility at the centre of a host of

other pathologies has been a popular model of pathophysiology for a variety of modern complex disease presentations, including autoimmune diseases. See Li and Flaws (2004).

10 For Wang Haogu's perspectives on the spleen and stomach, see *Wang Haogu yixue quanshu* 王好古醫學全書 (2004: 352ff.). For Li Shizhen's perspectives on the spleen and stomach, see Wang Xiaoping (2003). For a discussion of Li Gao's approach to gynaecology, see Chace (2001).

11 For an overview of the concept of *gewu zhizhi*, and the debates surrounding it, see Andrew Plaks' entry in the *RoutledgeCurzon Encyclopedia of Confucianism*: '*Gewu zhizhi* (Putting all things into the correct conceptual grid, extending to the utmost one's range of comprehension)' (Yao 2003, vol. 1: 226–7). On the influence of *gewu zhizhi* in pharmacology, see Métailié (2001).

12 For a further discussion of Zhu Zhenheng's intellectual legacy, see Furth (2006). For more on Zhu Zhenheng's approach to diseases of the stomach and spleen, see Ni and Damone (1992).

13 Both texts have been translated into English. See Yang (1993, 1994). For further discussion of scholar-physicians in the Jin-Yuan period, see Chen (1997); Ding *et al.* (1999); Leung (2001, 2003a, 2003b).

14 For an examination of itinerant doctors in Chinese medical history, see Qiu (2008); Unschuld (1979).

15 For an English language translation of the *Comprehensive Materia Medica*, see Luo (2003).

16 For a comprehensive collection of Li Shizhen's works, see *Li Shizhen yixue quanshu* 李时珍医学全书 (1999). For a biographical sketch of Li Shizhen, see Chace and Shima (2010: 11–18); Unschuld (1986: 145–8). For further discussion of Li Shizhen's *Comprehensive Materia Medica*, see Métailié (2001, 2010); Nappi (2009b).

17 For an excellent discussion of Li Shizhen's pulse diagnosis, see Li, Morris, Li and Mondot (2011).

18 For a translation of Li's *Exposition on the Eight Extraordinary Vessels*, see Chace and Shima (2010).

19 Li's organisation of this text, pieced together as it is from earlier writings, is also emblematic of a distinctive style of medical writing that developed during this time. For an examination of the influence of this syncretic style of medical writing on Ming dynasty medicine, see Simonis (2010). On inner alchemy, see Chapters 29 and 30 in this volume.

20 On these physiological terms and their relationship to inner alchemy, see Chapters 29 and 30 in this volume.

21 *Houtian jing* 後天精 is often rendered in English as 'postnatal essence', as it is produced from imbibed substances such as food, concentrated into blood, which is further concentrated into essence. For a detailed discussion of Zhang Jiebin's physiology of *mingmen*, and its importance for understandings of sexuality and reproduction, see Wu (2010, ch. 3).

22 For a discussion of the influence of the *Golden Mirror of Medicine* in late imperial China, see Hanson (2003).

23 For a comprehensive collection of case records reflecting the synthesis of these two currents in clinical practice, see Luo *et al.* (2004).

24 For an English language translation and commentary on the *Discourse on Warm Disease*, see Van Wart and Chace (2006).

25 The text is also known as the *Itemised Differentiation of Damp Heat* (*Shire tiaobian* 濕熱條辨).

26 An English language translation of *Systematic Differentiation of Warm Disease* in digital format is forthcoming. See Boyanton, Cm-db.com (Chinese Medicine Database).

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